Vivekanand Education Society's Institute of Technology

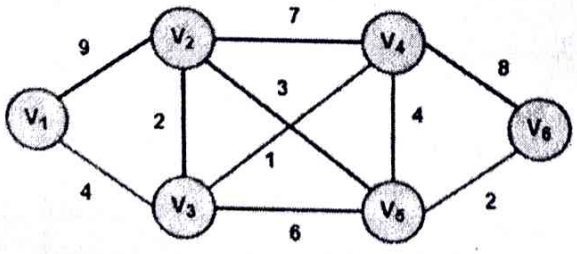
An Autonomous Institute Affiliated to University of Mumbai

End Semester Examination
Regular Exam - Summer 2025

Max marks: 60 | Duration: 2 hours
Branch: Computer Engineering | Semester: IV
Name of the Course: Artificial Intelligence | QP Code: R24-NCMMM41_032024-25
Course code:NCMMM41

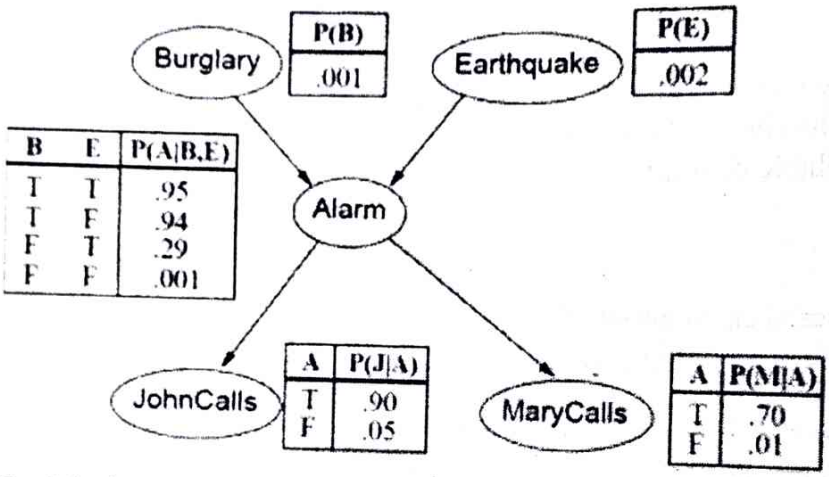
N.B. (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary.

		Marks
(a)	Explain different types of environments for Intelligent agents with examples. Describe PEAS for "Vacuum Cleaner world ".	10
(b)	Compare and contrast Supervised, Unsupervised, and Reinforcement Learning. Provide real-world applications for each type.	10
(a)	Explain and formulate the water jug problem. Write a state space representation for the same. Consider the starting state: both jugs are empty and generate the solution	10
(b)	What is planning in AI? Explain partial order planning and total order planning in detail.	10
(a)	Explain the alpha-beta pruning algorithm in detail. How does it improve the efficiency of the MiniMax algorithm? Support your answer with an example game tree and show the pruning steps clearly	10
(b)	Differentiate between Informed and uninformed search techniques. Apply UCS algorithms to reach the goal (node V6) from the start state (v1)	10



Q.4 (a) In the given Bayesian belief network, find out: 10

1. What is the probability that John calls?
2. What is the probability that the alarm has sounded but neither a burglary nor an earthquake occurred, and both John and Mary call?



(b) Explain the hill-climbing algorithm in detail with a suitable example. Discuss the problems associated with it. How to overcome these problems. 10

Q.5 (a) Apply forward chaining. "As per the law, it is a crime for an American to sell weapons to hostile nations. Country A, an enemy of America, has some missiles, and all the missiles were sold to it by Robert, who is an American citizen." Prove that "Robert is a criminal." 10

(b) Explain how propositional logic can be used to represent knowledge and make decisions in the Wumpus World. Illustrate your answer with suitable examples, including how an agent can infer the presence or absence of a pit or the Wumpus based on percepts 10

X-----X-----X----X

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